

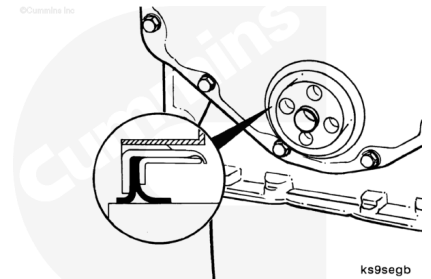
Trainings ▾

General Information

Lip Seal

Front gear train engines use a lip style front crankshaft Seal (shown) in which the rotating portion of the sealing occurs at the contact surface between the lip of the seal and the crankshaft.

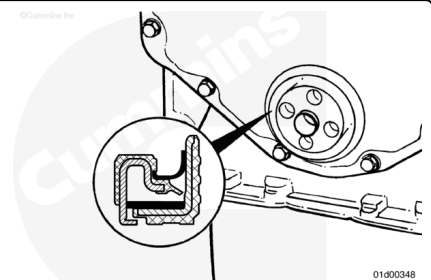
Always replace the front crankshaft seal with the same style seal as was previously installed.



Unitized Seal

Rear gear train engines use a dual or non-lip style seal which utilize a built in wear sleeve and a concealed sealing lip. The inner and out diameter are press-fit onto the crankshaft and the front gear cover respectively, requiring service tool, Part Number 3164659, to remove and install. The sealing point is internal to the seal.

Always replace the front crankshaft seal with the same style seal as was previously installed.

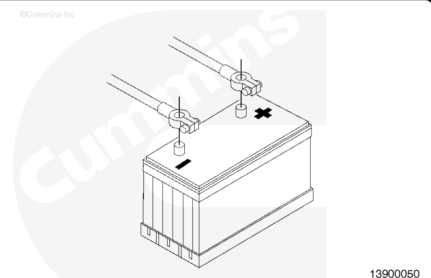


Preparatory Steps

Front Gear Train

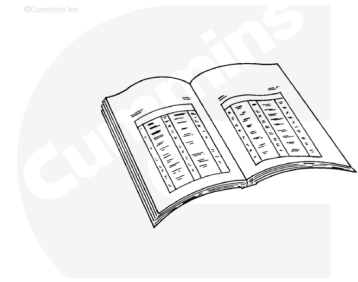
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Disconnect the batteries.

- Remove the engine water pump (cooling fan) drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)
- Remove the fan hub. Refer to Procedure 008-039 (</qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html>)
- Remove the vibration damper/crankshaft pulley. If equipped with a viscous damper, Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>), If equipped with a rubber damper, Refer to Procedure 001-051 (</qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html>). If equipped with a crankshaft pulley only, Refer to Procedure 001-022 (</qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html>)
- Remove the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html>).

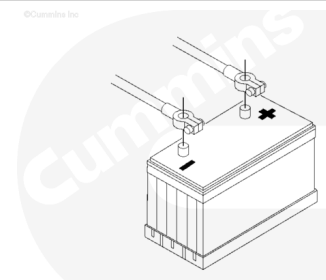


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Rear Gear Train

⚠ WARNING ⚠

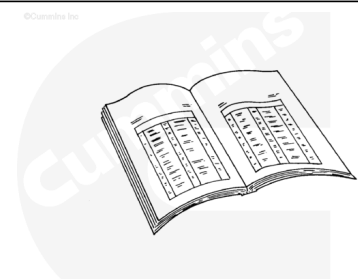
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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Disconnect the batteries.

- Remove the cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)
- Remove the vibration damper/crankshaft pulley. If equipped with a viscous damper, Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>). If equipped with a rubber damper, Refer to Procedure 001-051 (</qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html>). If equipped with a crankshaft pulley only, Refer to



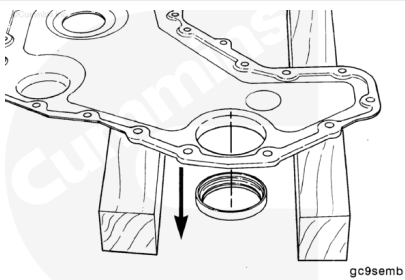
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Remove

Front Gear Train

Support the front gear cover on a flat work surface with wooden blocks. Using a suitable punch and hammer, drive the old seal out of the front gear cover from the back side of the cover to the front side.

Note : Some engines may have an addition dust seal installed in front of the front crankshaft seal.



Rear Gear Train

Use tool, Part Number 3164659, to remove the front crankshaft seal from the front gear cover.

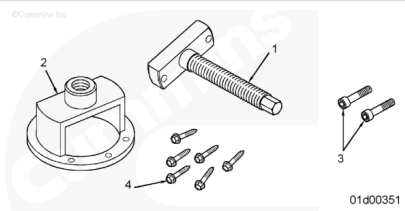


Table 1.
Front
Crankshaft
Seal
Replacer Kit,
Part Number
3164659

Item Number	Part Number	Description	Quantity
1	3164667	Replace screw assembly	1
2	3164661	Crankshaft seal replacer	1
3	3164239	Socket head capscrew, M12 x 1.25 x 60 mm	2

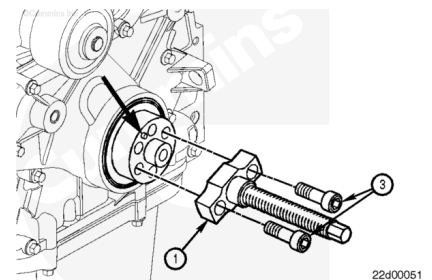
Table 1.
Front
Crankshaft
Seal
Replacer Kit,
Part Number
3164659

Item Number	Part Number	Description	Quantity
4	3164217	Sheet metal screw, Number 10 x 25.4 mm [1 in] long	25 (6 shown)
Not shown	3164218	Drill. 3.57 mm [9/64 in]	1

Note : The front gear cover does not need to be removed to remove and install the crankshaft seal.

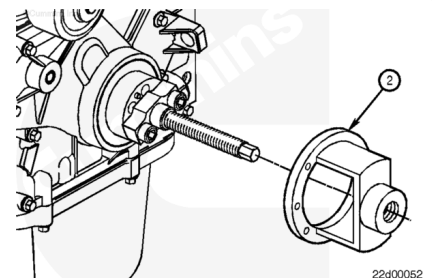
Mount the replacer screw assembly (1) onto the crankshaft nose.

Install the two M12 x 1.25 x 60-mm socket head capscrews (3).



Lubricate the replacer screw with anti-seize or a suitable grease.

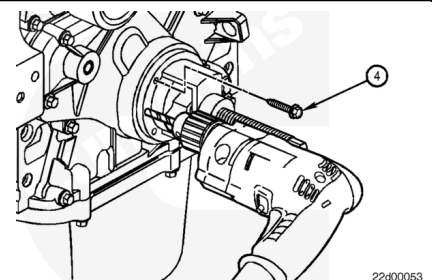
Hold the replacer screw and install the crankshaft seal replacer (2) onto the replacer screw assembly. Advance the crankshaft seal replacer toward the seal by rotating it **clockwise** until it is positioned against the front gear cover.



⚠ CAUTION ⚠

Drill the hole carefully and straight to reduce the possibility of damage to the front cover or the crankshaft.

Service Tip: Because of space restrictions, it may be necessary to use a compact right angle drill. Also, it may be necessary to shorten the drill bit used to drill the sheet metal



screw holes.

Note : The flange of the crankshaft seal replacer is 8 mm [0.32 in] thick.

Mark the drill to a depth of 18 mm [0.71 in] with tape for drill depth control and apply grease to the drill to catch the chips. Stop frequently to remove the chips.

With the crankshaft seal replacer positioned against the front gear cover, drill one hole 10 mm [0.39 in] deep. Make sure the marking tape has **not** moved from the original position.

Install one sheet metal screw (4) into the seal to hold the crankshaft seal replacer in position.

Drill five additional holes 10 mm [0.39 in] deep and install the additional five sheet metal screws.

Note : Make sure all six sheet metal screws are threaded through both the inner and outer seal casings. The sheet metal screws need to be uniformly tightened in order to pull out the inner and outer casings of the seal together.

⚠ CAUTION ⚠

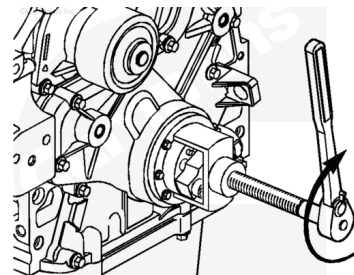
Do not use an impact wrench or air tools. Doing so can damage the tool.

Slowly rotate the replacer screw **clockwise** until the seal is removed. Do **not** exceed torque value.

Torque Value: 33 n•m [45 ft-lb]

Note : If the sheet metal screws pull out of the seal or **only** the inner casing pulls out, stop the removal operation. Rotate the replacer screw **counterclockwise** to force the inner casing back. Remove the sheet metal screws, slightly reorient the tool, drill new holes, and install the sheet metal screws in the new locations.

Complete the removal procedure, remove the tool, and discard the old seal.



Clean and Inspect for Reuse

⚠ WARNING ⚠

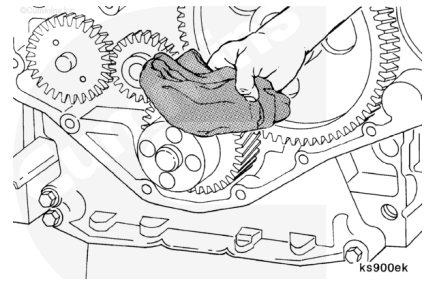
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Using solvent, clean the oil and seal residue from the crankshaft surface.

Dry with compressed air.

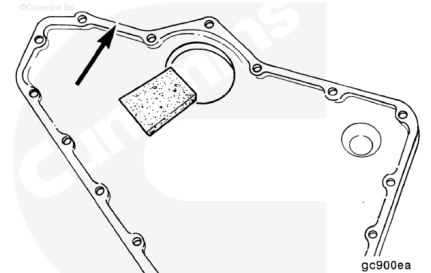
For front gear train engines, use a gasket scraper or abrasive pad, Part Number 3823258 or equivalent, to remove all sealant on the front gear cover and front gear housing.

Clean the gasket sealing surfaces and the surface between the oil seal and front gear cover.

Use solvent to clean the front gear cover.

For rear gear train engines, clean the front gear cover seal bore of any seal residue. Inspect the front gear cover seal bore for nicks or burrs. Use an abrasive pad, Part Number 3823258 or equivalent, to remove any minor damage.

Dry with compressed air.

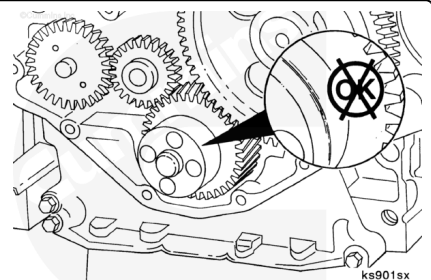


⚠ CAUTION ⚠

The seal lip and the sealing surface on the crankshaft must be free from all oil residue to prevent seal leaks.

Note : If the crankshaft has excessive wear, a service wear sleeve is available for engines that use a lip style front crankshaft seal.

Inspect the nose of the crankshaft for excessive wear.



For engines equipped with a non-lip style front crankshaft seal, use a fine crocus cloth to remove any nicks or burrs. No wear sleeve is available if the crankshaft nose sealing surface is damaged.

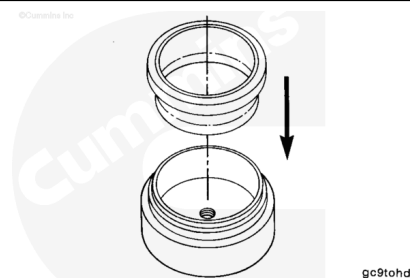
For engines equipped with a lip style front crankshaft seal, inspect the crankshaft seal contact area for a wear groove. If the groove is deeper than 0.25 mm [0.010 in], a wear sleeve and oversize seal **must** be used. Refer to Procedure 001-025 (/qs3/pubsys2/xml/en/procedures/40/40-001-025-tr.html).

Install

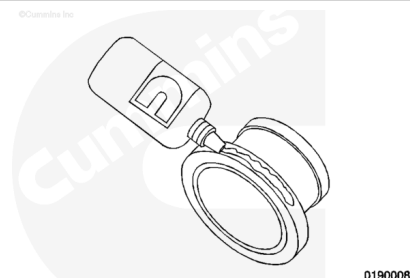
Front Gear Train

Leave the plastic pilot installation tool in the front crankshaft seal.

Position the seal on the seal installation tool, Part Number 3824498, with the seal dust lip facing outward.



Apply a bead of sealant, Part Number 3164070, to the outside circumference of the seal.



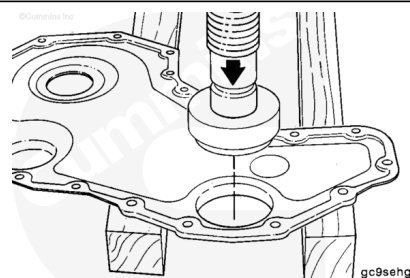
⚠ CAUTION ⚠

Properly support the front cover lubricating oil seal flange to prevent damage to the lubricating oil seal and front cover.

Press the lubricating oil seal into the front cover from the backside of the cover toward the frontside of the cover.

Press the lubricating oil seal until the oil seal installation tool, Part Number 3824498, bottoms against the front cover.

Note : Each front crankshaft seal kit comes with an installation tool. This tool can be used in place of front



crankshaft seal installation tool, Part No. 3824498, if not available. Using a mallet, tap around the installation tool to drive the front crankshaft seal into the front gear cover until the installation tool bottoms against the front gear cover.

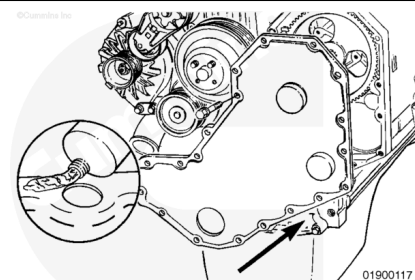
The seal lip and the sealing surface on the crankshaft **must** be free from all oil residue to prevent seal leaks.

Apply a thin bead of sealant, Part Number 3164070, to the sealing surface of the front cover.

Note : Do not remove the plastic seal pilot tool from the front crankshaft seal at this time. Use the plastic seal pilot tool to guide the seal on the crankshaft.

Note : Install the front cover within 10 minutes of applying the sealant, or the sealant will not seal correctly. Once installed, allow the sealant to dry for 30 minutes before running the engine.

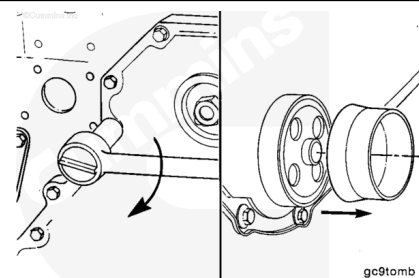
Install the front gear cover on the engine, using the plastic seal pilot tool to guide the front crankshaft seal onto the crankshaft.



Install and tighten the front gear cover capscrews.

Torque Value: 24 n·m [18 ft·lb]

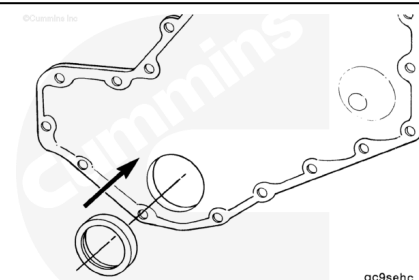
Remove the plastic pilot tool from the crankshaft.



If previously equipped or if the engine operates in a dusty environment, install a dust seal.

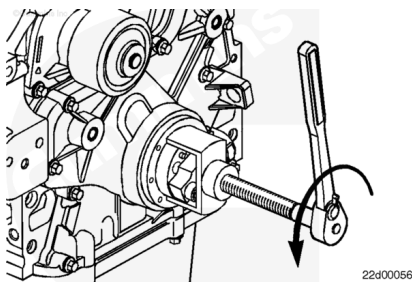
A pilot tool is not necessary for the dust seal. Slide the dust seal over the nose of the crankshaft.

Use the installation tool that came with the new front crankshaft seal to install the dust seal into the front gear cover. Install the dust seal until it is even with the front gear cover.



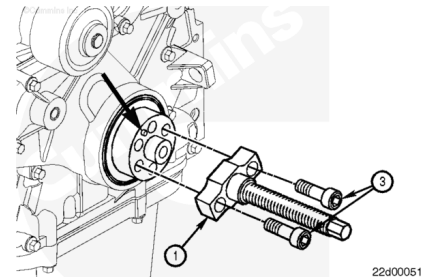
Rear Gear Train

Use tool, Part Number 3164659, to install the oil seal into the front gear cover.



Mount the replacer screw assembly (1) onto the crankshaft nose.

Install the two M12 x 1.25 x 60 mm socket head capscrews (3).



⚠ CAUTION ⚠

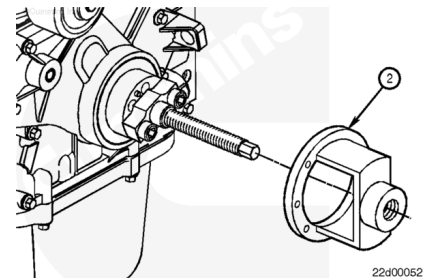
Do not use an impact wrench or air tools. Doing so can damage the tool.

Place the new front crankshaft seal over the crankshaft nose and slide it by hand toward the front gear cover as far as possible.

Note : Make sure the seal is positioned squarely with the crankshaft.

While holding the replacer screw, install the crankshaft seal replacer (2) onto the replacer screw assembly.

Advance the crankshaft seal replacer toward the seal by rotating it **clockwise** until it is positioned against the seal.

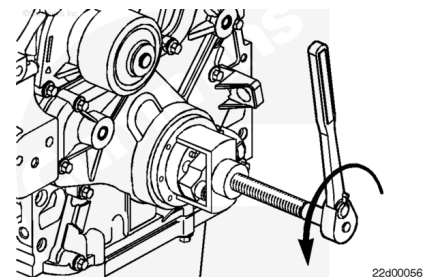


⚠ CAUTION ⚠

Do not overtighten the replacer screw assembly after the crankshaft seal replacer contacts the front cover. Doing so can damage the tool.

While holding the crankshaft seal replacer, rotate the replacer screw **counterclockwise** until the crankshaft seal replacer contacts the front gear cover.

Remove the service tools.



Finishing Steps

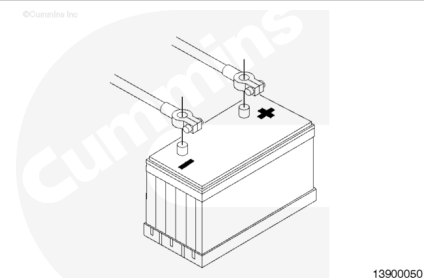
Front Gear Train

- Install the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html>)
- Install the vibration damper/crankshaft pulley. If equipped with a viscous damper, Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>). If equipped with a rubber damper, Refer to Procedure 001-051 (</qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html>). If equipped with a crankshaft pulley only, Refer to Procedure 001-022 (</qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html>)
- Install the fan hub if removed. Refer to Procedure 008-039 (</qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html>)
- Install the engine water pump (cooling fan) drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>).



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Connect the batteries.

Operate the engine and check for leaks.

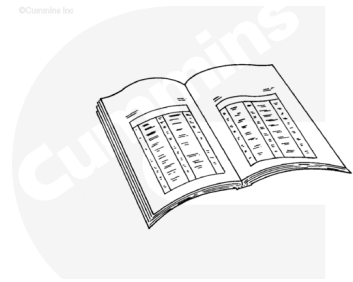
Rear Gear Train

- Install the vibration damper pulley/crankshaft pulley. If equipped with a viscous damper, Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>). If equipped with a rubber damper, Refer to

Procedure 001-051

(/qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html). If equipped with a crankshaft pulley only, refer to Refer to Procedure Procedure 001-022 (/qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html)

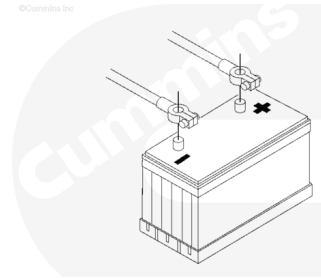
- Install the cooling fan drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html).



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Connect the batteries.

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